Grade 6 · Unit 1 · Ecosystems and their Interactions — Flashcards

PA STEELS 3.1.6-8.F, I, J, K, L, U; 3.4.6-8.C

GR 6 · UNIT 1 | STEELS 3.1.6-8.I | #1

Ecosystem

All the living things in an area plus the nonliving things they interact with, such as water, soil, and sunlight.

Gr 6 · Unit 1 · #1 · 3.1.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 | STEELS 3.1.6-8.I | #2

Biotic factor

A living or once-living part of an ecosystem. Examples: plants, animals, fungi, bacteria, fallen leaves.

Gr 6 · Unit 1 · #2 · 3.1.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 | STEELS 3.1.6-8.I | #3

Abiotic factor

A nonliving part of an ecosystem. Examples: sunlight, water, air, soil, temperature.

Gr 6 · Unit 1 · #3 · 3.1.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 | STEELS 3.1.6-8.I | #4

Organism

A single living thing. Example: one oak tree or one deer.

Gr 6 · Unit 1 · #4 · 3.1.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 | STEELS 3.1.6-8.J | #5

Species

A group of organisms that are alike and can reproduce and have young that can also reproduce. Example: white-tailed deer.

Gr 6 · Unit 1 · #5 · 3.1.6-8.J

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 | STEELS 3.1.6-8.I | #6

Population

All the organisms of ONE species living in the same area. Example: all the deer in a forest.

Gr 6 · Unit 1 · #6 · 3.1.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 | STEELS 3.1.6-8.I | #7

Community

All the populations of DIFFERENT species living in the same area.

Gr 6 · Unit 1 · #7 · 3.1.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 | STEELS 3.1.6-8.I | #8

Resource

Something an organism needs to live and grow, such as food, water, oxygen, space, or sunlight.

Gr 6 · Unit 1 · #8 · 3.1.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 | STEELS 3.1.6-8.J | #9

Interdependent

Depending on each other. Organisms in an ecosystem depend on other living things and on nonliving things to survive.

Gr 6 · Unit 1 · #9 · 3.1.6-8.J

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 | STEELS 3.1.6-8.I | #10

What happens to populations when resources are plentiful or scarce?

When resources are plentiful, organisms grow and populations increase. When resources are scarce, growth slows and populations shrink.

Gr 6 · Unit 1 · #10 · 3.1.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 | STEELS 3.1.6-8.I | #11

Competition

When organisms that need the same limited resources (food, water, space) struggle to get them. It limits their growth and reproduction.

Gr 6 · Unit 1 · #11 · 3.1.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.J #12

Predation

One organism hunts, kills, and eats another. It can reduce the size of a population or even wipe it out.

Gr 6 · Unit 1 · #12 · 3.1.6-8.J

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.J #13

Predator

An organism that hunts and eats other organisms. Example: a hawk.

Gr 6 · Unit 1 · #13 · 3.1.6-8.J

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.J #14

Prey

An organism that is hunted and eaten. Example: a mouse.

Gr 6 · Unit 1 · #14 · 3.1.6-8.J

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.J #15

Mutualism

A mutually beneficial interaction: BOTH organisms are helped. Some partners depend on each other so much that neither can survive alone. Example: bees and flowers.

Gr 6 · Unit 1 · #15 · 3.1.6-8.J

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.J #16

Do the same kinds of interactions happen in every ecosystem?

Yes. The species are different in every ecosystem, but the same kinds of interactions (competition, predation, mutualism) happen in all of them.

Gr 6 · Unit 1 · #16 · 3.1.6-8.J

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.F #17

Photosynthesis

Plants, algae, and many microorganisms use energy from light to make sugar (food) from carbon dioxide and water. Oxygen is released.

Gr 6 · Unit 1 · #17 · 3.1.6-8.F

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.F #18

Algae

Simple, plant-like organisms, mostly in water, that make their own food by photosynthesis. Examples: seaweed, pond scum.

Gr 6 · Unit 1 · #18 · 3.1.6-8.F

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.F #19

Microorganism

A living thing too small to see without a microscope. Examples: bacteria and many algae.

Gr 6 · Unit 1 · #19 · 3.1.6-8.F

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.F #20

Phytoplankton

Tiny algae and other microorganisms that float in water and make food by photosynthesis.

Gr 6 · Unit 1 · #20 · 3.1.6-8.F

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.K #21

Producer

An organism that makes its own food, usually by photosynthesis. Examples: plants, algae.

Gr 6 · Unit 1 · #21 · 3.1.6-8.K

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.K #22

Consumer

An organism that gets energy and matter by eating other organisms. Examples: rabbits, hawks.

Gr 6 · Unit 1 · #22 · 3.1.6-8.K

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.K #23

Decomposer

An organism that breaks down dead plants and animals and recycles their nutrients back to the soil or water. Examples: fungi, bacteria.

Gr 6 · Unit 1 · #23 · 3.1.6-8.K

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.K #24

Nutrient

A substance that living things need to live and grow. Decomposers return nutrients to the soil and water.

Gr 6 · Unit 1 · #24 · 3.1.6-8.K

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.K #25

Food chain

One path that matter and energy follow as one organism eats another. Example: grass → rabbit → fox.

Gr 6 · Unit 1 · #25 · 3.1.6-8.K

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.K #26

Food web

A model of many connected food chains. Shows how matter and energy move between producers, consumers, and decomposers.

Gr 6 · Unit 1 · #26 · 3.1.6-8.K

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.K #27

How does energy move through an ecosystem?

Energy enters most ecosystems as sunlight, passes through the food web, and leaves as heat. Energy flows through; it is not recycled.

Gr 6 · Unit 1 · #27 · 3.1.6-8.K

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.K #28

What happens to matter in an ecosystem?

The atoms that make up organisms are used again and again, moving between living and nonliving parts of an ecosystem. Matter is not used up.

Gr 6 · Unit 1 · #28 · 3.1.6-8.K

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.K #29

System

A group of parts that work together. Scientists choose the boundaries of a system, such as one pond, to track the matter and energy moving in and out.

Gr 6 · Unit 1 · #29 · 3.1.6-8.K

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.K #30

Atom

The tiny particle that all matter is made of. The atoms in living things are used again and again.

Gr 6 · Unit 1 · #30 · 3.1.6-8.K

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.K #31

Conservation of matter

Matter is not created or destroyed. In an ecosystem it changes form and moves between living and nonliving parts, and into or out of the ecosystem, but none is lost.

Gr 6 · Unit 1 · #31 · 3.1.6-8.K

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.K #32

Terrestrial

Living on or having to do with land. Example: a forest is a terrestrial ecosystem.

Gr 6 · Unit 1 · #32 · 3.1.6-8.K

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.K #33

Aquatic

Living in or having to do with water. Example: a pond is an aquatic ecosystem.

Gr 6 · Unit 1 · #33 · 3.1.6-8.K

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.L #34

What happens when part of an ecosystem changes?

The sizes of populations can change. Example: a drought, a new disease, or a new predator can make a population shrink.

Gr 6 · Unit 1 · #34 · 3.1.6-8.L

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.L #35

Empirical evidence

Information gathered by observing or measuring, such as data from an experiment or field study.

Gr 6 · Unit 1 · #35 · 3.1.6-8.L

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.U #36

Biodiversity

The variety of species in an ecosystem. It is often used as a measure of an ecosystem's health.

Gr 6 · Unit 1 · #36 · 3.1.6-8.U

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.U #37

Design solution

A plan or invention made to solve a problem. Example: planting trees along a stream to protect water quality.

Gr 6 · Unit 1 · #37 · 3.1.6-8.U

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.U #38

Ecosystem services

Benefits people get from healthy ecosystems. Examples: cleaning water, recycling nutrients, preventing soil erosion.

Gr 6 · Unit 1 · #38 · 3.1.6-8.U

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 STEELS 3.1.6-8.U #39

Criteria

What a design solution must do to be successful. Example: keep a stream clean enough for fish.

Gr 6 · Unit 1 · #39 · 3.1.6-8.U

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 | STEELS 3.1.6-8.U | #40

Constraints

Limits on a design solution. They can be scientific, economic (cost), or social (what people want or accept).

Gr 6 · Unit 1 · #40 · 3.1.6-8.U

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 | STEELS 3.4.6-8.C | #41

Watershed

An area of land where all the water drains into the same stream, river, or lake. Example: Reading is in the Schuylkill River watershed.

Gr 6 · Unit 1 · #41 · 3.4.6-8.C

✂ cut on dashes · fold on dots

GR 6 · UNIT 1 | STEELS 3.4.6-8.C | #42

Wetland

Land that is covered by water for at least part of the year, such as a marsh or swamp. Wetlands filter water, soak up floodwater, and are home to many species.

Gr 6 · Unit 1 · #42 · 3.4.6-8.C

✂ cut on dashes · fold on dots

Grade 6 · Unit 2 · Weather and Climate Systems — Flashcards

PA STEELS 3.3.6-8.H, I, J, O

GR 6 · UNIT 2 | STEELS 3.3.6-8.H | #1

Water cycle

Water continually moves among the land, ocean, and atmosphere, changing between solid, liquid, and gas. Also called the hydrologic cycle.

Gr 6 · Unit 2 · #1 · 3.3.6-8.H

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 | STEELS 3.3.6-8.H | #2

Atmosphere

The layer of air that surrounds Earth.

Gr 6 · Unit 2 · #2 · 3.3.6-8.H

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 | STEELS 3.3.6-8.H | #3

State of matter

The form matter is in: solid, liquid, or gas. In the water cycle, water changes state between ice, liquid water, and water vapor.

Gr 6 · Unit 2 · #3 · 3.3.6-8.H

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.H #4

Water vapor

Water in the form of a gas. It is invisible and is part of the air.

Gr 6 · Unit 2 · #4 · 3.3.6-8.H

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.H #5

What drives the water cycle?

Energy from the sun (which evaporates water) and the force of gravity (which pulls rain down and water downhill).

Gr 6 · Unit 2 · #5 · 3.3.6-8.H

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.H #6

Gravity

The force that pulls objects toward Earth. In the water cycle, gravity pulls rain and snow down and makes water flow downhill.

Gr 6 · Unit 2 · #6 · 3.3.6-8.H

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.H #7

Evaporation

Liquid water changes into water vapor (a gas) when it is heated, mostly by the sun.

Gr 6 · Unit 2 · #7 · 3.3.6-8.H

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.H #8

Transpiration

Plants release water vapor into the air through their leaves.

Gr 6 · Unit 2 · #8 · 3.3.6-8.H

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.H #9

Condensation

Water vapor cools and changes into tiny liquid droplets. This is how clouds form.

Gr 6 · Unit 2 · #9 · 3.3.6-8.H

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.H #10

Crystallization (water)

Water vapor or liquid water changes into solid ice crystals, such as when snowflakes form in a cloud.

Gr 6 · Unit 2 · #10 · 3.3.6-8.H

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.H #11

Precipitation

Water that falls from clouds to the ground as rain, snow, sleet, or hail. Gravity pulls it down.

Gr 6 · Unit 2 · #11 · 3.3.6-8.H

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 | STEELS 3.3.6-8.H | #12

Runoff

Water that flows downhill over land into streams, rivers, and lakes.

Gr 6 · Unit 2 · #12 · 3.3.6-8.H

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 | STEELS 3.3.6-8.J | #13

Weather

The condition of the air at one place and time: temperature, air pressure, humidity, precipitation, and wind.

Gr 6 · Unit 2 · #13 · 3.3.6-8.J

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 | STEELS 3.3.6-8.I | #14

Climate

The usual pattern of weather in a region over many years.

Gr 6 · Unit 2 · #14 · 3.3.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 | STEELS 3.3.6-8.J | #15

Air mass

A huge body of air that has about the same temperature and humidity throughout.

Gr 6 · Unit 2 · #15 · 3.3.6-8.J

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.J #16

Air pressure

The push of the air pressing down on a surface. It can be high or low.

Gr 6 · Unit 2 · #16 · 3.3.6-8.J

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.J #17

Humidity

The amount of water vapor in the air.

Gr 6 · Unit 2 · #17 · 3.3.6-8.J

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.J #18

Wind

Air moving from an area of HIGH pressure to an area of LOW pressure.

Gr 6 · Unit 2 · #18 · 3.3.6-8.J

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.J #19

Front

The boundary where two different air masses meet. When air masses collide, the weather can change suddenly, often with storms.

Gr 6 · Unit 2 · #19 · 3.3.6-8.J

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.J #20

Weather forecast

A prediction of the weather. Weather is so complex that it can only be predicted as a chance (probability), like "70% chance of rain."

Gr 6 · Unit 2 · #20 · 3.3.6-8.J

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.I #21

Why is the equator warmer than the poles?

Sunlight hits the equator more directly than the poles, so the equator is warmer. This uneven heating drives winds and ocean currents.

Gr 6 · Unit 2 · #21 · 3.3.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.I #22

Latitude

How far north or south a place is from the equator. Places farther from the equator are usually colder.

Gr 6 · Unit 2 · #22 · 3.3.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.I #23

Altitude

How high a place is above sea level. Higher places are usually colder.

Gr 6 · Unit 2 · #23 · 3.3.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.I #24

Prevailing winds

Winds that usually blow from the same direction in a region. They form in bands caused by sunlight heating Earth unevenly.

Gr 6 · Unit 2 · #24 · 3.3.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.I #25

Rotation (Earth)

Earth spinning on its axis, once about every 24 hours. It causes day and night and the Coriolis effect.

Gr 6 · Unit 2 · #25 · 3.3.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.I #26

Coriolis effect

Earth's rotation makes winds and ocean currents curve instead of moving in straight lines.

Gr 6 · Unit 2 · #26 · 3.3.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.I #27

Ocean current

A large flow of moving ocean water. Currents carry heat around the globe. They are driven by winds and by differences in water temperature and saltiness (density).

Gr 6 · Unit 2 · #27 · 3.3.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.I #28

Salinity

How much salt is dissolved in water. Saltier water is denser.

Gr 6 · Unit 2 · #28 · 3.3.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.I #29

Density

How much mass is packed into a space. Denser water (colder or saltier) sinks below less dense water.

Gr 6 · Unit 2 · #29 · 3.3.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.I #30

Convection

Movement in a liquid or gas caused by differences in temperature and density: warm material rises and cooler material sinks. In the ocean, it moves heat around the globe.

Gr 6 · Unit 2 · #30 · 3.3.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.I #31

How does the ocean affect climate?

The ocean absorbs energy from the sun, releases it slowly, and moves it around the world. This makes coastal climates milder.

Gr 6 · Unit 2 · #31 · 3.3.6-8.I

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.O #32

Global warming

The rise in Earth's average surface temperature over the past century. Human activities are a major cause.

Gr 6 · Unit 2 · #32 · 3.3.6-8.O

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.O #33

Greenhouse gas

A gas in the atmosphere that traps heat. Examples: carbon dioxide and methane.

Gr 6 · Unit 2 · #33 · 3.3.6-8.O

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.O #34

Carbon dioxide

A gas in the air (CO_2). Plants use it for photosynthesis. Burning fossil fuels adds more of it, which traps more heat.

Gr 6 · Unit 2 · #34 · 3.3.6-8.O

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.O #35

Methane

A greenhouse gas (CH_4) that traps heat. It comes from sources such as farm animals, landfills, and natural gas.

Gr 6 · Unit 2 · #35 · 3.3.6-8.O

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.O #36

Fossil fuels

Coal, oil, and natural gas. Burning them releases carbon dioxide, the main human cause of global warming.

Gr 6 · Unit 2 · #36 · 3.3.6-8.O

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.O #37

What natural causes can change Earth's climate?

Changes in the amount of sunlight reaching Earth and gases from volcanic eruptions can also change global temperatures.

Gr 6 · Unit 2 · #37 · 3.3.6-8.O

✂ cut on dashes · fold on dots

GR 6 · UNIT 2 STEELS 3.3.6-8.O #38

Solar radiation

Energy from the sun that reaches Earth, including light and heat.

Gr 6 · Unit 2 · #38 · 3.3.6-8.O

✂ cut on dashes · fold on dots

Grade 6 · Unit 3 · Earth's Dynamic Systems — Flashcards

PA STEELS 3.3.6-8.D, E, F, G, L

GR 6 · UNIT 3 | STEELS 3.3.6-8.F | #1

Rock cycle

Earth's materials are constantly changed and recycled into new rocks by melting, crystallization, weathering, deformation, and sedimentation.

Gr 6 · Unit 3 · #1 · 3.3.6-8.F

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 | STEELS 3.3.6-8.F | #2

What energy drives the rock cycle?

Energy from the sun (weathering and erosion at the surface) and from Earth's hot interior (melting and moving rock deep underground).

Gr 6 · Unit 3 · #2 · 3.3.6-8.F

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 | STEELS 3.3.6-8.F | #3

Melting (rocks)

Deep underground, rock gets hot enough to turn into liquid.

Gr 6 · Unit 3 · #3 · 3.3.6-8.F

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.F #4

Crystallization (rocks)

Melted rock cools and hardens, and minerals form crystals.

Gr 6 · Unit 3 · #4 · 3.3.6-8.F

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.F #5

Mineral

A natural, nonliving solid with a crystal structure. Rocks are made of minerals.

Gr 6 · Unit 3 · #5 · 3.3.6-8.F

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.E #6

Weathering

The breaking down of rock into smaller pieces by water, ice, wind, and living things.

Gr 6 · Unit 3 · #6 · 3.3.6-8.E

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.E #7

Erosion

The moving of weathered rock and soil from one place to another by water, ice, wind, or gravity.

Gr 6 · Unit 3 · #7 · 3.3.6-8.E

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.E #8

Deposition

When water, wind, or ice slows down and drops the sediment it was carrying in a new place.

Gr 6 · Unit 3 · #8 · 3.3.6-8.E

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.F #9

Sediment

Small pieces of rock, sand, mud, and once-living material that are carried and dropped by water, wind, or ice.

Gr 6 · Unit 3 · #9 · 3.3.6-8.F

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.F #10

Sedimentation

Layers of sediment settle, pile up, and slowly harden into rock.

Gr 6 · Unit 3 · #10 · 3.3.6-8.F

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.F #11

Deformation

Rock bends, folds, or breaks because of pushing forces from moving plates.

Gr 6 · Unit 3 · #11 · 3.3.6-8.F

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.E #12

How do caves form?

Water moving underground can dissolve rock and create caves. Example: Crystal Cave in Berks County.

Gr 6 · Unit 3 · #12 · 3.3.6-8.E

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.E #13

Groundwater

Water that soaks into the ground and moves through soil and cracks in rock. It can dissolve rock and form caves.

Gr 6 · Unit 3 · #13 · 3.3.6-8.E

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.E #14

What are some slow changes to Earth's surface?

Some processes change the surface over millions of years. Examples: plates moving, mountains being pushed up, rivers carving valleys.

Gr 6 · Unit 3 · #14 · 3.3.6-8.E

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.E #15

Uplift

When forces inside Earth slowly push land upward. Uplift forms mountain ranges.

Gr 6 · Unit 3 · #15 · 3.3.6-8.E

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.E #16

Catastrophic event

A sudden event that changes Earth's surface quickly. Examples: earthquake, volcanic eruption, landslide, meteor impact.

Gr 6 · Unit 3 · #16 · 3.3.6-8.E

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.E #17

Landslide

Rock and soil suddenly sliding down a slope.

Gr 6 · Unit 3 · #17 · 3.3.6-8.E

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.G #18

Tectonic plate

A huge slab of Earth's outer layer. Plates move slowly, a few centimeters a year, and have moved great distances over time.

Gr 6 · Unit 3 · #18 · 3.3.6-8.G

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.G #19

Mid-ocean ridge

An underwater mountain range where plates spread apart and new seafloor forms.

Gr 6 · Unit 3 · #19 · 3.3.6-8.G

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.G #20

Ocean trench

A deep valley in the seafloor where old seafloor sinks back down into Earth and is destroyed.

Gr 6 · Unit 3 · #20 · 3.3.6-8.G

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.G #21

Fracture zone

A long scar in the seafloor that crosses a mid-ocean ridge, left where pieces of plates slid past each other.

Gr 6 · Unit 3 · #21 · 3.3.6-8.G

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.G #22

Continental shelf

The shallow underwater edge of a continent. Including the shelves makes the continents fit together even better.

Gr 6 · Unit 3 · #22 · 3.3.6-8.G

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.G #23

What evidence shows that plates move?

Matching fossils and rock types on continents far apart, continents whose shapes fit together like puzzle pieces, and ridges and trenches on the seafloor.

Gr 6 · Unit 3 · #23 · 3.3.6-8.G

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.D #24

Rock layers (strata)

Layers of rock stacked on top of each other. They record events in Earth's history.

Gr 6 · Unit 3 · #24 · 3.3.6-8.D

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.D #25

Relative age

Whether a rock or fossil is older or younger than another. In undisturbed layers, the oldest layers are at the bottom and the youngest are at the top.

Gr 6 · Unit 3 · #25 · 3.3.6-8.D

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.D #26

Absolute age

The actual age of something in years. Rock layers and fossils alone give only relative ages, not absolute ages.

Gr 6 · Unit 3 · #26 · 3.3.6-8.D

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.D #27

Fossil

The preserved remains or traces of an organism that lived long ago, usually found in rock layers.

Gr 6 · Unit 3 · #27 · 3.3.6-8.D

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.D #28

Geologic time scale

A timeline that organizes Earth's history into major events using evidence from rock layers and fossils.

Gr 6 · Unit 3 · #28 · 3.3.6-8.D

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.D #29

How old is Earth?

About 4.6 billion years old.

Gr 6 · Unit 3 · #29 · 3.3.6-8.D

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.L #30

Natural hazard

A natural event that can harm people and property. Examples: earthquakes, volcanic eruptions, floods, severe storms.

Gr 6 · Unit 3 · #30 · 3.3.6-8.L

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.L #31

Magnitude

How big or strong a natural event is. Example: an earthquake's magnitude measures how much energy it released.

Gr 6 · Unit 3 · #31 · 3.3.6-8.L

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.L #32

Frequency (hazards)

How often a natural hazard happens in a place. Mapping past events helps forecast where and how likely future ones are.

Gr 6 · Unit 3 · #32 · 3.3.6-8.L

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.E #33

Earthquake

Sudden shaking of the ground when rocks underground break and move.

Gr 6 · Unit 3 · #33 · 3.3.6-8.E

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.E #34

Volcano

An opening in Earth's surface where melted rock, ash, and gases erupt.

Gr 6 · Unit 3 · #34 · 3.3.6-8.E

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.L #35

Mass wasting

Rock and soil moving downhill because of gravity, such as landslides and mudslides.

Gr 6 · Unit 3 · #35 · 3.3.6-8.L

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.L #36

Tsunami

A series of huge ocean waves, usually caused by an earthquake or volcanic eruption under the sea.

Gr 6 · Unit 3 · #36 · 3.3.6-8.L

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.L #37

Drought

A long period with much less rain than usual. Reservoirs that store water can reduce its effects.

Gr 6 · Unit 3 · #37 · 3.3.6-8.L

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.L #38

Which natural hazards can be predicted?

Some hazards, like volcanic eruptions and severe weather, have warning signs that allow predictions. Earthquakes happen suddenly and cannot yet be predicted.

Gr 6 · Unit 3 · #38 · 3.3.6-8.L

✂ cut on dashes · fold on dots

GR 6 · UNIT 3 STEELS 3.3.6-8.L #39

Hazard mitigation

Actions and technologies that reduce damage from natural hazards. Examples: early-warning systems, earthquake-safe buildings, flood walls.

Gr 6 · Unit 3 · #39 · 3.3.6-8.L

✂ cut on dashes · fold on dots